

US00PP34883P2

(12) **United States Plant Patent**
Halladin

(10) **Patent No.:** **US PP34,883 P2**
(45) **Date of Patent:** **Jan. 3, 2023**

(54) **PICEA PUNGENS TREE NAMED ‘INMOHOOP’**

(50) Latin Name: *Picea pungens*
Varietal Denomination: **INMoHoop**

(71) Applicant: **Early Morning LLC**, Boring, OR (US)

(72) Inventor: **Paul H. Halladin**, Boring, OR (US)

(73) Assignee: **EARLY MORNING LLC**, Boring, OR (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/875,954**

(22) Filed: **Jul. 28, 2022**

(51) **Int. Cl.**
A01H 7/00 (2006.01)

(52) **U.S. Cl.**
USPC **Plt./213**

(58) **Field of Classification Search**
USPC Plt./213
CPC A01H 7/00; A01H 5/00
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

PP5,457 P * 4/1985 Holden Plt./213
* cited by examiner

Primary Examiner — June Hwu
(74) Attorney, Agent, or Firm — Weatherly IP Solutions, LLC; Barbara Campbell

(57) **ABSTRACT**

A new and distinct variety of *Picea pungens* tree having an orthotropic leader growth, a dwarf and wide conical shape, a compact habit, blue-colored needles/foliage, and long and thin needles, as disclosed.

3 Drawing Sheets

1

Genus and species: *Picea pungens*.
Denomination: ‘INMoHoop’.

BACKGROUND

‘INMoHoop’ is a new and distinct variety of *Picea pungens*. ‘INMoHoop’ originated from a controlled cross conducted in May 2001 in a greenhouse in Boring, Oreg. of the female *Picea pungens* parent ‘Montgomery’(unpatented) and the female *Picea pungens* parent ‘Hoopsi’(unpatented). The seeds from this cross germinated in May 2002.

‘INMOHOOP’ was selected and first asexually propagated in January 2009 in Boring, Oreg. via grafting onto *Picea abies* rootstock (unpatented).

‘INMoHoop’ reproduces true to type in successive generations since 2009 of asexual reproduction via grafting onto *Picea abies* rootstock (unpatented).

SUMMARY

The following are the most outstanding and distinguishing characteristics of this new variety when grown under normal horticultural practices in Boring, Oreg.

1. Orthotropic leader growth;
2. Dwarf and wide conical shape;
3. Compact habit;
4. Blue-colored needles/foliage; and
5. Long and thin needles.

DESCRIPTION OF THE PHOTOGRAPHS

This new *Picea pungens* tree is illustrated by the accompanying photographs which show the plant’s overall plant habit including form and foliage. The photographs are of a 10-year-old tree grown outdoors in Boring, Oreg. and were taken in June 2022 under cloudy conditions. The colors

2

shown are as true as can be reasonably obtained by conventional photographic procedures.

FIG. 1 shows the overall form of the whole tree of ‘INMoHoop’.

FIG. 2 shows a close-up of the buds and needles or leaves of the tree of FIG. 1.

FIG. 3 shows a close-up of the terminal branch shoot of the tree of FIG. 1.

DETAILED DESCRIPTION

The following detailed descriptions set for the distinctive characteristics of ‘INMoHoop’. The data which define these characteristics were collected outdoors in Boring, Oreg. in June 2022 on 10-year-old trees in Boring, Oreg. Color references are to The R.H.S. Colour Chart of The Royal Horticultural Society of London (R.H.S.) 2001, except where general terms of ordinary dictionary significance are used.

Classification:

Family.—Pinaceae.
Botanical.—*Picea pungens*.
Denomination.—‘INMoHoop’.

Tree:

Type.—Evergreen conifer.
Shape.—Wide conical.
Growth and branching habit.—Dwarf compact tree, single flush.
Propagation type.—Grafting onto *Picea abies* rootstock (unpatented).
Height (from soil line).—150 cm at 10 years.
Width (horizontal plant diameter).—120 cm at 10 years.
Length of terminal 1-year shoots.—20.0 cm to 30.0 cm.

Length of subterminal 1-year shoots.—10.0 cm to 15.0 cm.
Diameter of trunk.—10.0 cm at 10 years at a height of 30.0 cm on the trunk.
Temperature tolerance or zones.—USDA Zones 2 to 8.
Trunk surface texture.—Surface has irregular, small, and rough scaly plates.
Trunk color.—RHS N199D (Greyed-Brown) and RHS N167D (Greyed-Orange).
Lateral branch color for 1-year-old shoot.—RHS 166D (Greyed-Orange).
Leaves (needles):
Arrangement.—Radial, single.
Quantity of leaves/needles per terminal one-year branch shoot.—450 to 475.
Quantity of leaves/needles per subterminal one-year branch shoot.—200 to 250.
Shape.—Needle-like, acicular.
Apex.—Acute.
Base.—Attenuate, attached to leaf peg.
Margin.—Entire, smooth.
Leaf length, terminal shoot.—Basal: 1.5 cm to 2.0 cm; Medial: 3.0 cm to 3.5 cm.
Distal.—2.0 cm to 2.5 cm.
Width.—0.1 cm.
Texture.—Smooth and covered with waxy layer.
Pubescence.—Absent.
Curvature of the needles.—On the apical needles, there is no curvature (the needles are straight); on the terminal leaf branch shoot, the upper shoot needles have a very slight forward curvature and the needles on the side and lowers are straight (no curvature); on the small, lateral shoots, the curvature of the needles is absent to very slight.
Immature leaf color.—RHS 122D (Blue-Green).
Mature leaf color for both upper and lower surfaces.—RHS 122D (Blue-Green).
Mature leaf color when wax rubbed off.—RHS 189A (Greyed-Green).
Petioles.—Leaves are sessile on a short peg.
Buds:
Shape.—Conical with a pointed apex.
Color.—RHS 165C (Greyed-Orange).
Size.—2.0 cm height and 1.0 cm width at the leader branches and 1.2 cm height, 0.5 cm width at the terminal branches and 0.5 cm height and 0.3 cm width on the small secondary shoots.
Scales.—Present, smooth and papery and adhering to the bud shape when wet or newly developed; when dry or mature, the papery bud scales are reflexed from the tip.
Terminal bud burst.—Within the months of April and May.

Cones: No male or female cones have been observed to date.
Disease/pest resistance: No specific disease or pest resistance has been observed to date.

COMPARISON WITH PARENTAL AND COMMERCIAL LINES

Table 1 shows a comparison of the differences between ‘INMoHoop’ and the commercial *Picea pungens* tree ‘Fat Albert’ (unpatented).

TABLE 1

Comparison with Commercial Line		
Characteristic	‘INMoHoop’	‘Fat Albert’
Growth habit	Dwarf compact tree	Large open tree
Size, height × width at 10-years-old	150 cm × 120 cm	290 cm × 190 cm
Mature needle/leaf color	RHS 122D (Blue-Green)	RHS 130D (Green)

Table 2 shows a comparison of differences between ‘INMoHoop’ and the parental lines.

TABLE 2

Comparison with Parental Lines			
Characteristic	‘INMoHoop’	Male parent ‘Hoopsi’	Female parent ‘Montgomery’
Shape	Wide conical as both an immature and mature tree	Irregular conical	Irregular flattened mound as an immature tree and wide conical as a mature tree
Branch growth type	Orthotropic	Plagiotropic	Plagiotropic
Growth habit	Dwarf and compact	Large open tree; staking and pruning required for tree-like habit	Dwarf compact tree; flattened mound when immature and wide-conical when mature
Growth rate cm/year	20 cm to 30 cm	35 cm to 45 cm	10 cm to 20 cm
Size, height × width at 10-years-old	150 cm × 120 cm	230 cm × 135 cm	90 cm × 105 cm
Mature needle/leaf color	RHS 122D (Blue-Green)	RHS 112C (Blue-Green)	RHS 122B (Blue-Green)
Leaf/needle length, medial	3.0 cm to 3.5 cm	2.2 cm to 2.5 cm	2.2 cm to 2.5 cm
Leaf/needle width	0.1 cm	0.2 cm	0.1 cm

I claim:
1. A new and distinct variety of *Picea pungens* tree designated ‘INMoHoop’ as illustrated and described herein.

* * * * *



FIG. 1



FIG. 2

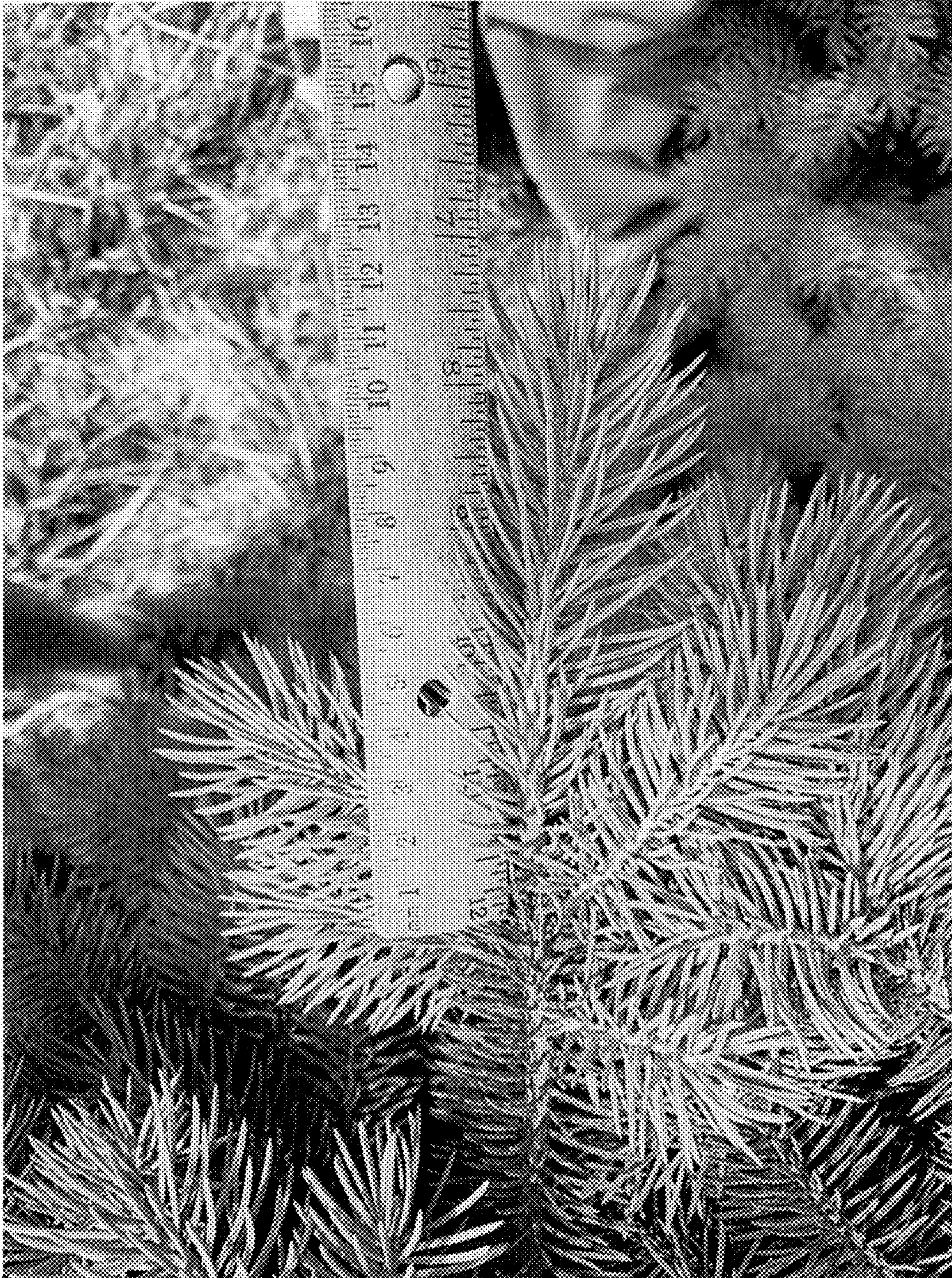


FIG. 3

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : PP34,883 P2
APPLICATION NO. : 17/875954
DATED : January 3, 2023
INVENTOR(S) : Paul H. Halladin

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

On the Title Page

In Column 1, Line 10 should read:
and the male *Picea pungens* parent 'Hoopsi' (unpatented).

Signed and Sealed this
Twenty-first Day of February, 2023


Katherine Kelly Vidal
Director of the United States Patent and Trademark Office